2025

Case #1

Risk Management

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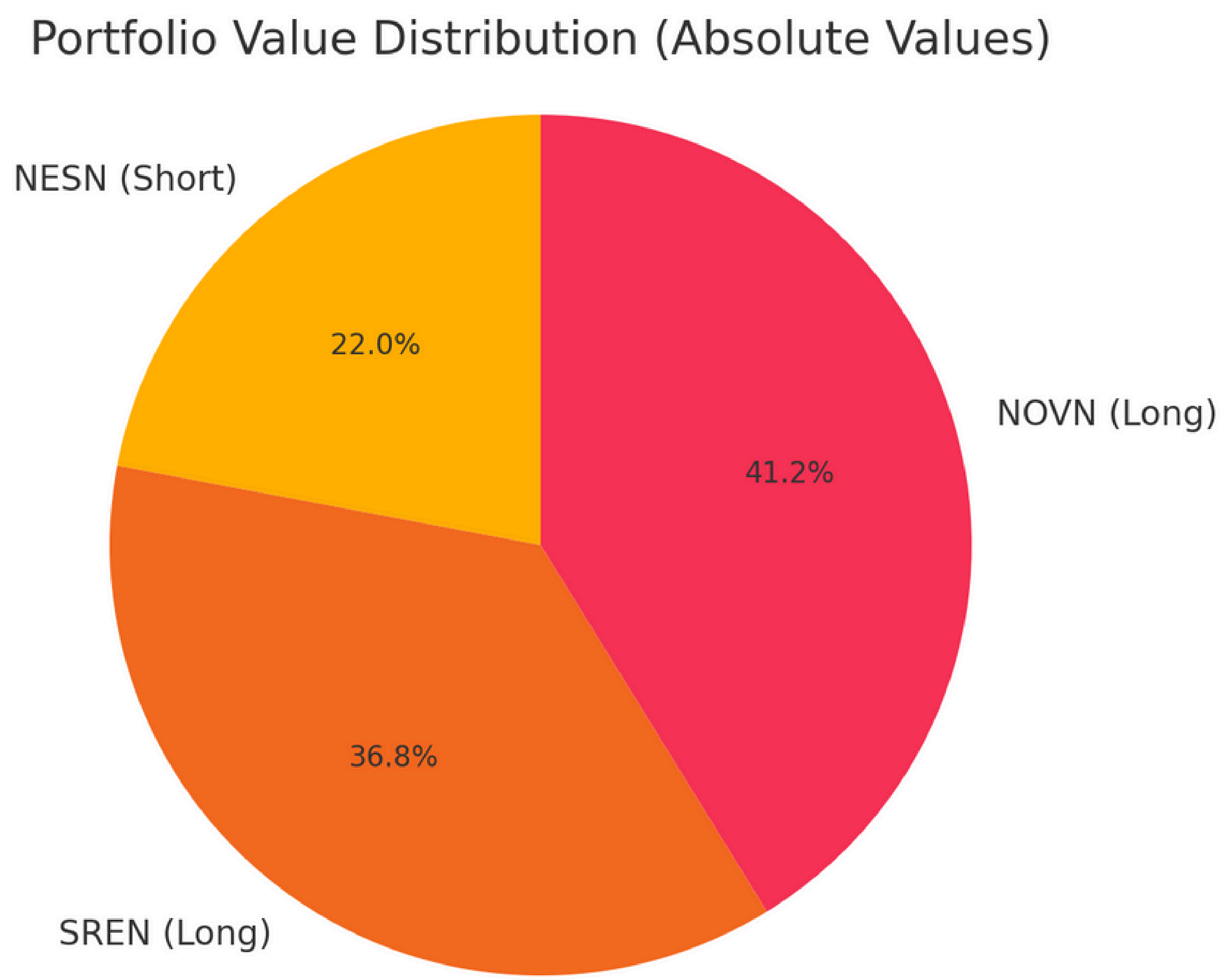
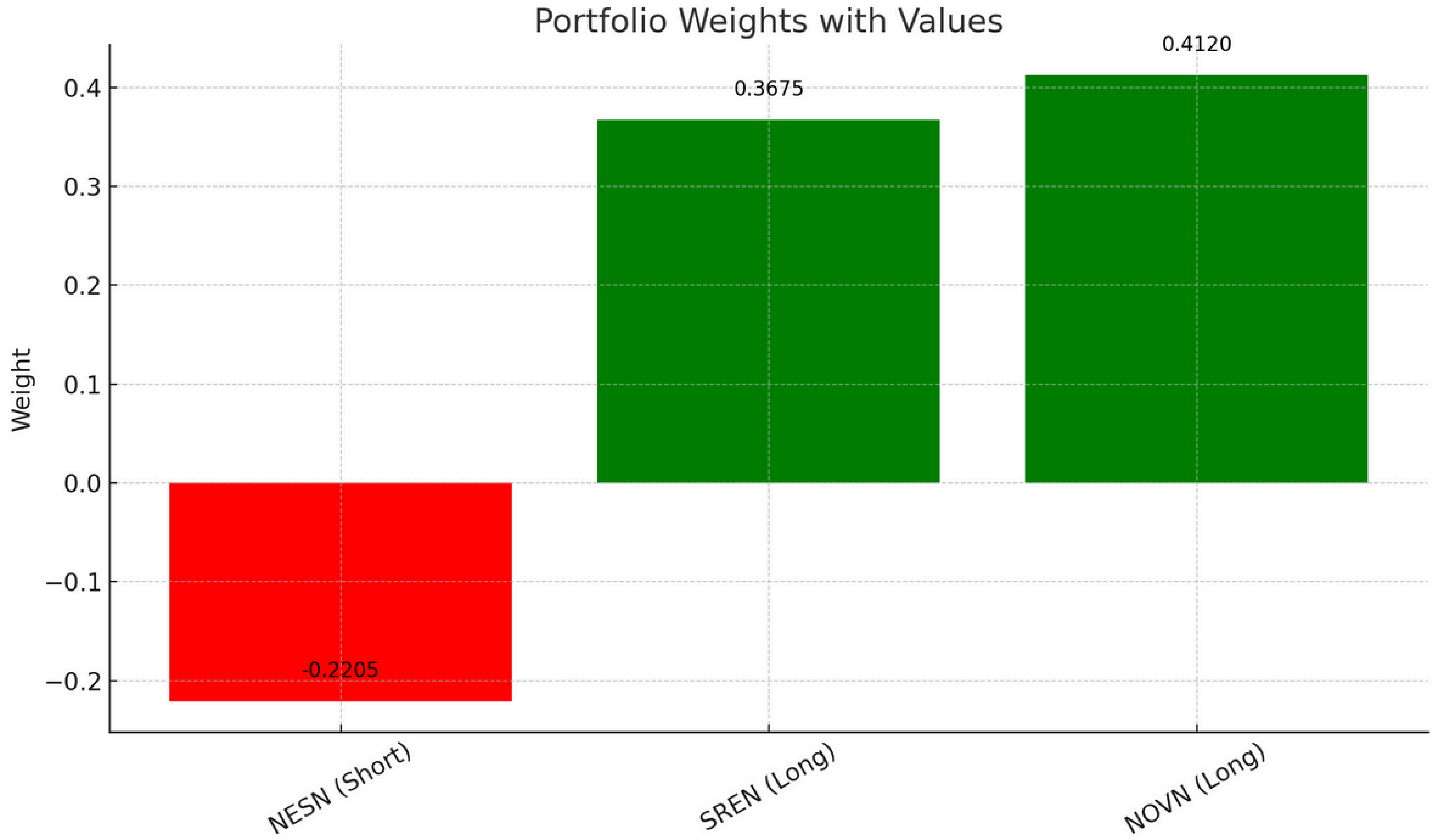
Part I

Portfolio Description



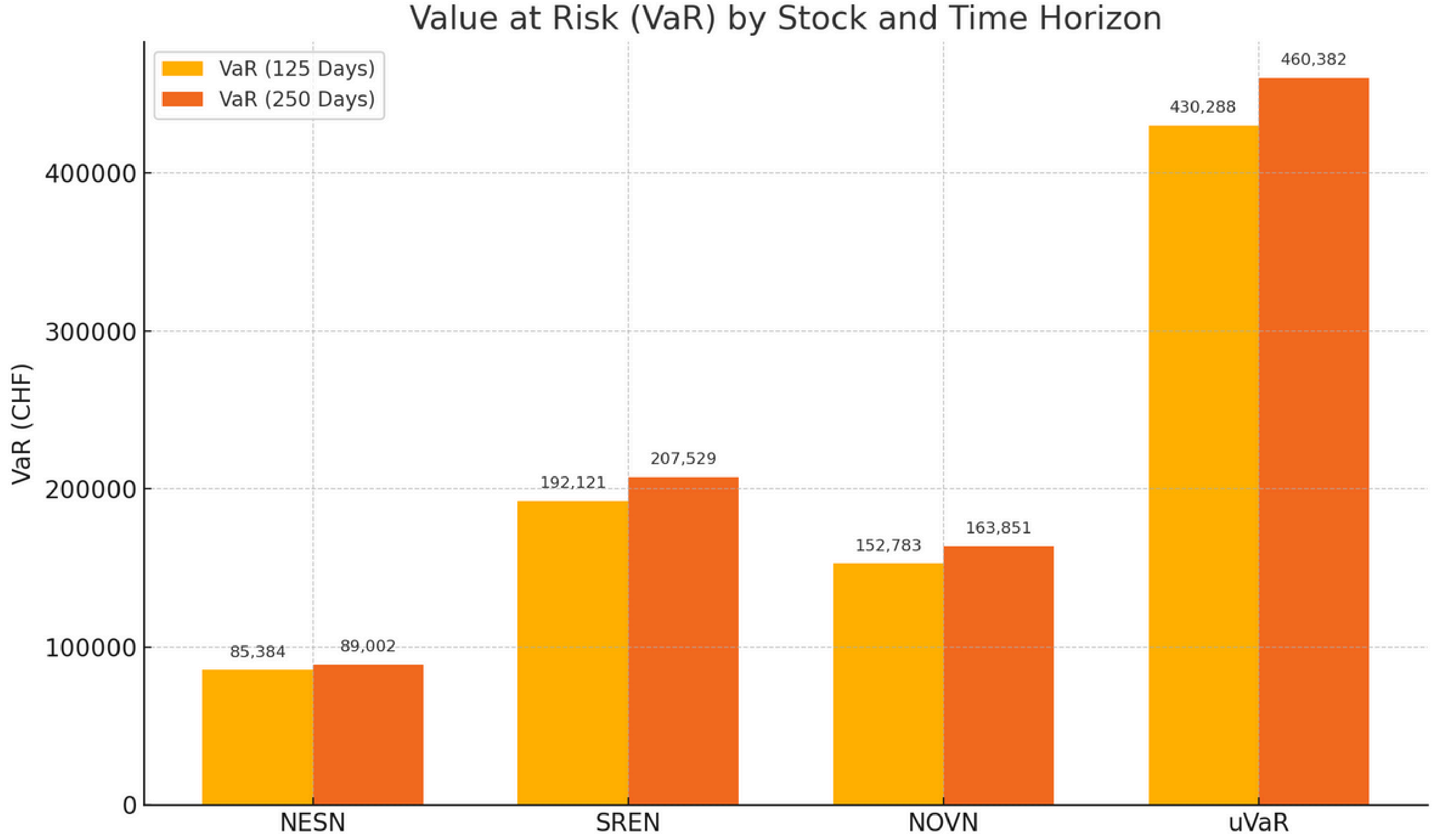


- formed on 21st November 2022 with the following positions:
  - **NESN**: 112.14 CHF x 30’000 shares
  - **SREN**: 80.12 CHF x 70’000 shares
  - **NOVN**: 78.60 CHF x 80’000 shares

Q1: Individual Equity VaR Analysis Using Historical Volatility

- 250-day window smooths short-term shocks but may underestimate recent risks.
- 125-day window is more responsive, hence more sensitive to volatility.
- Assumes i.i.d. returns:
  - Ignores volatility clustering and fat tails.
  - May underestimate risk in real markets, especially during turbulence.

$$\mathrm{VaR}_{99\%,1} = \alpha \cdot \sigma_{\text{stock}} \cdot \text{Position Value}$$



- Historical VaR is a useful baseline, but should be complemented by models capturing dynamic volatility patterns.

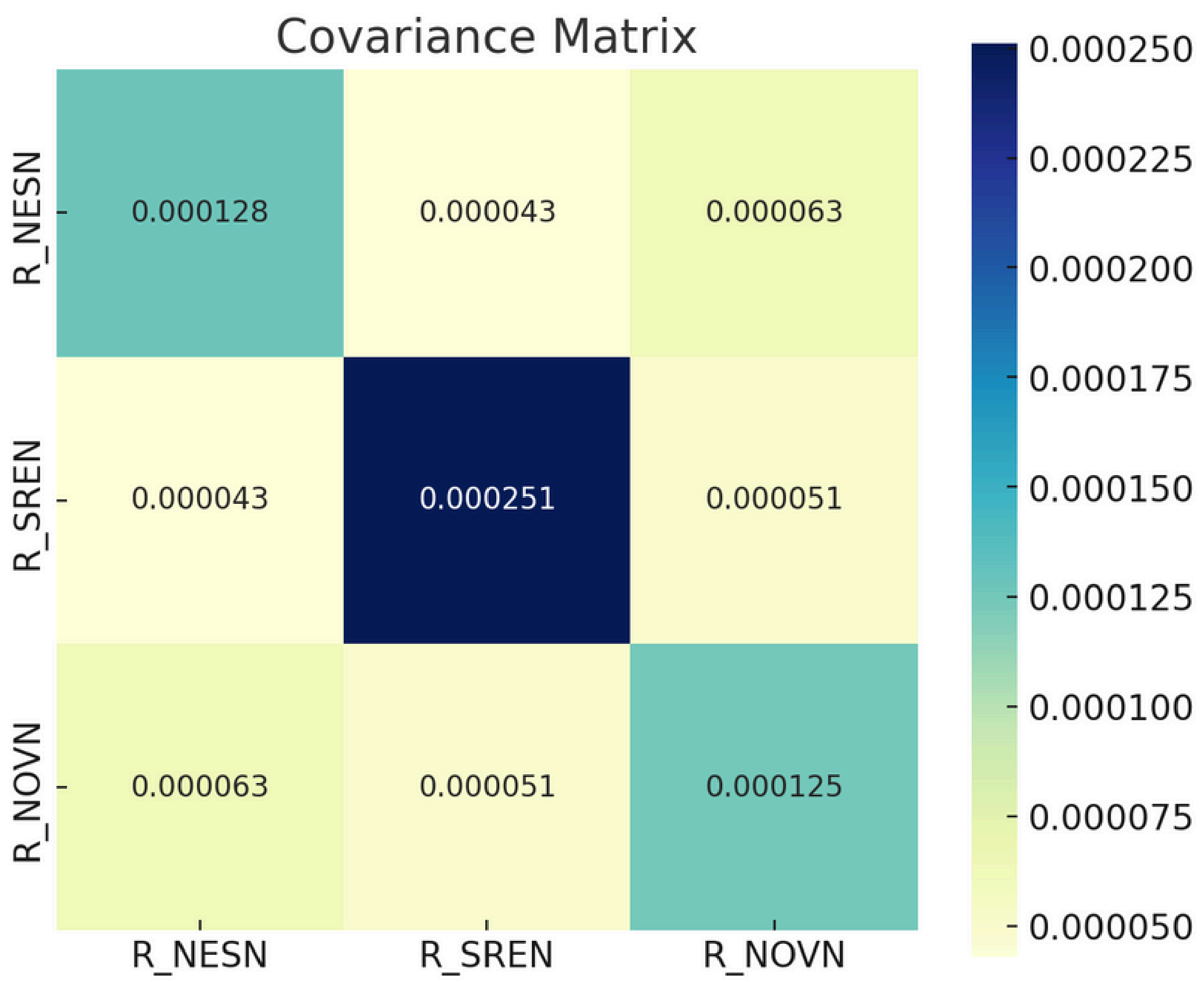
Q2: Portfolio VaR Using Asset-Normal Approach

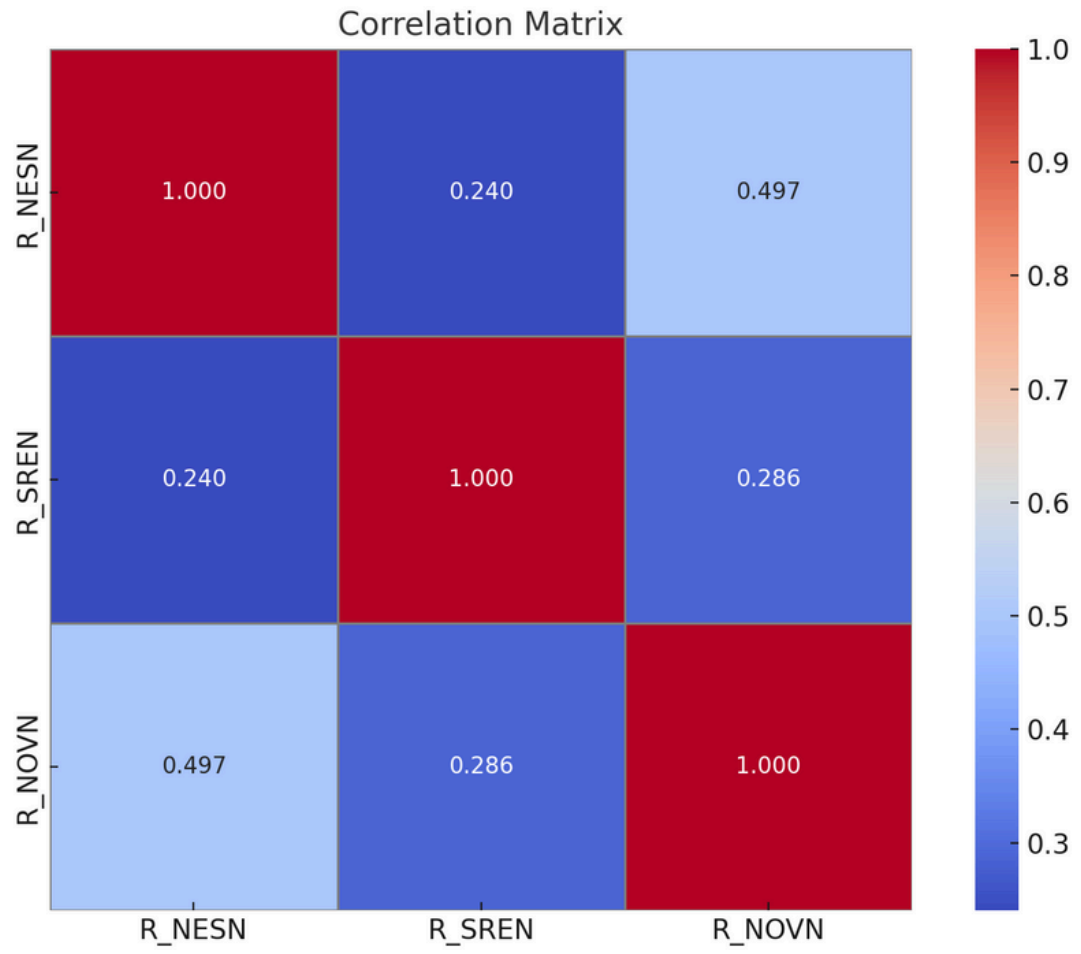
Methodology:

- computed correlation and covariance matrices to get the variance of the portfolio

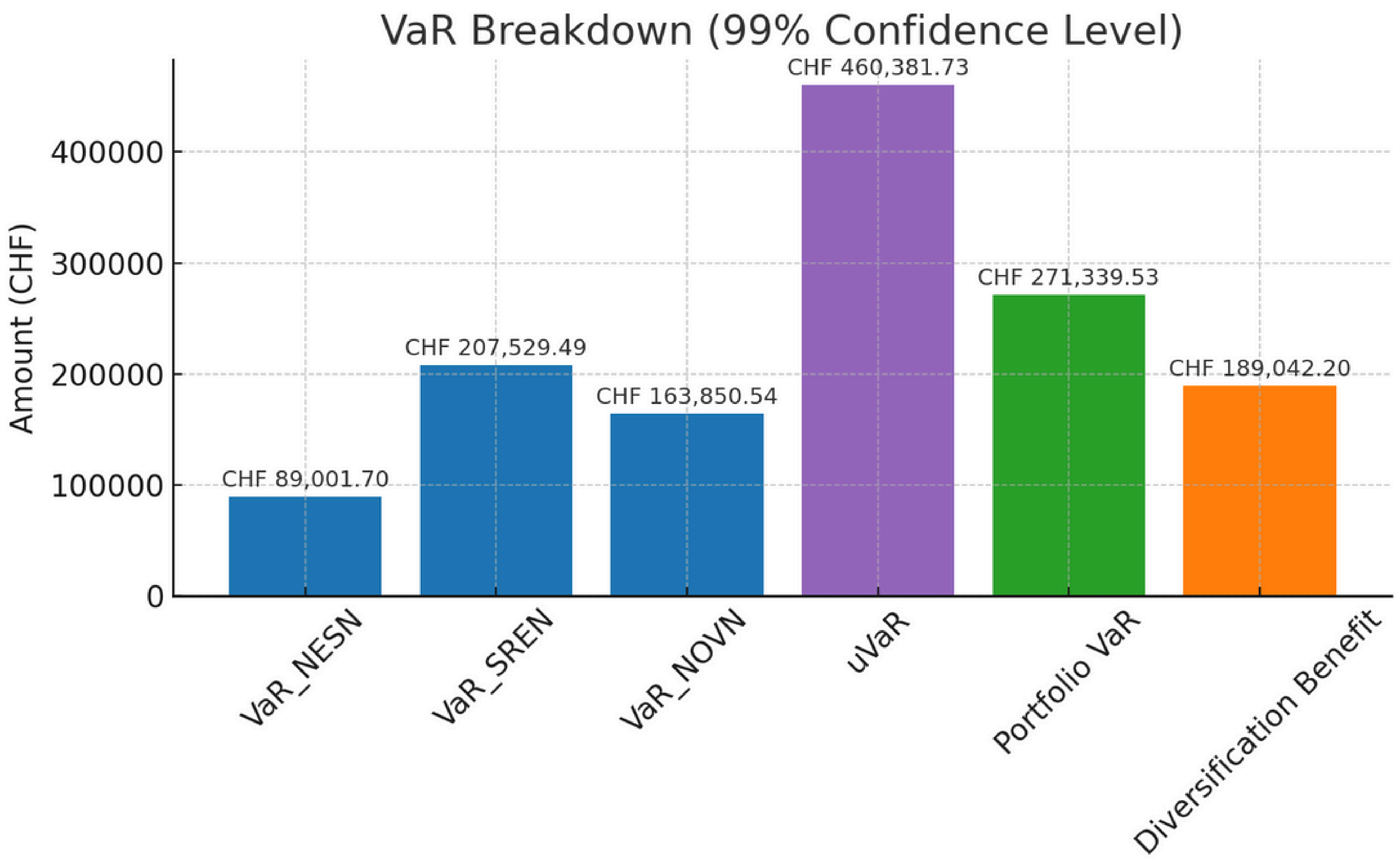
$$\sigma^2_{\text{portfolio}} = \mathbf{w}^T \Sigma \mathbf{w}$$

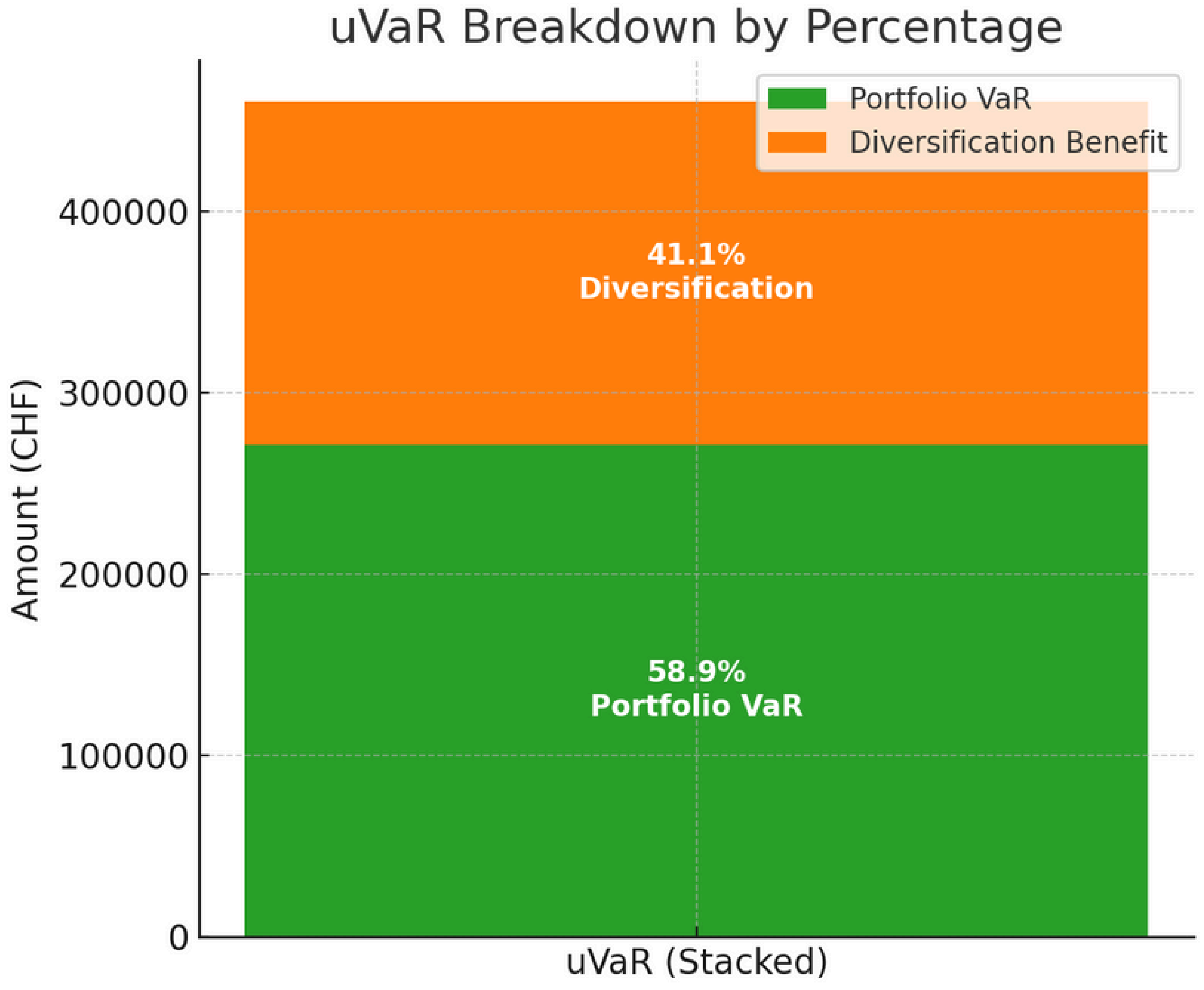
- assumed fixed number of shares
- applied the volatility in $\text{VaR}_{99\%,1} = \alpha \cdot \sigma_{\text{portfolio}} \cdot \text{Portfolio Value}$





Q2: Portfolio VaR Using Asset-Normal Approach

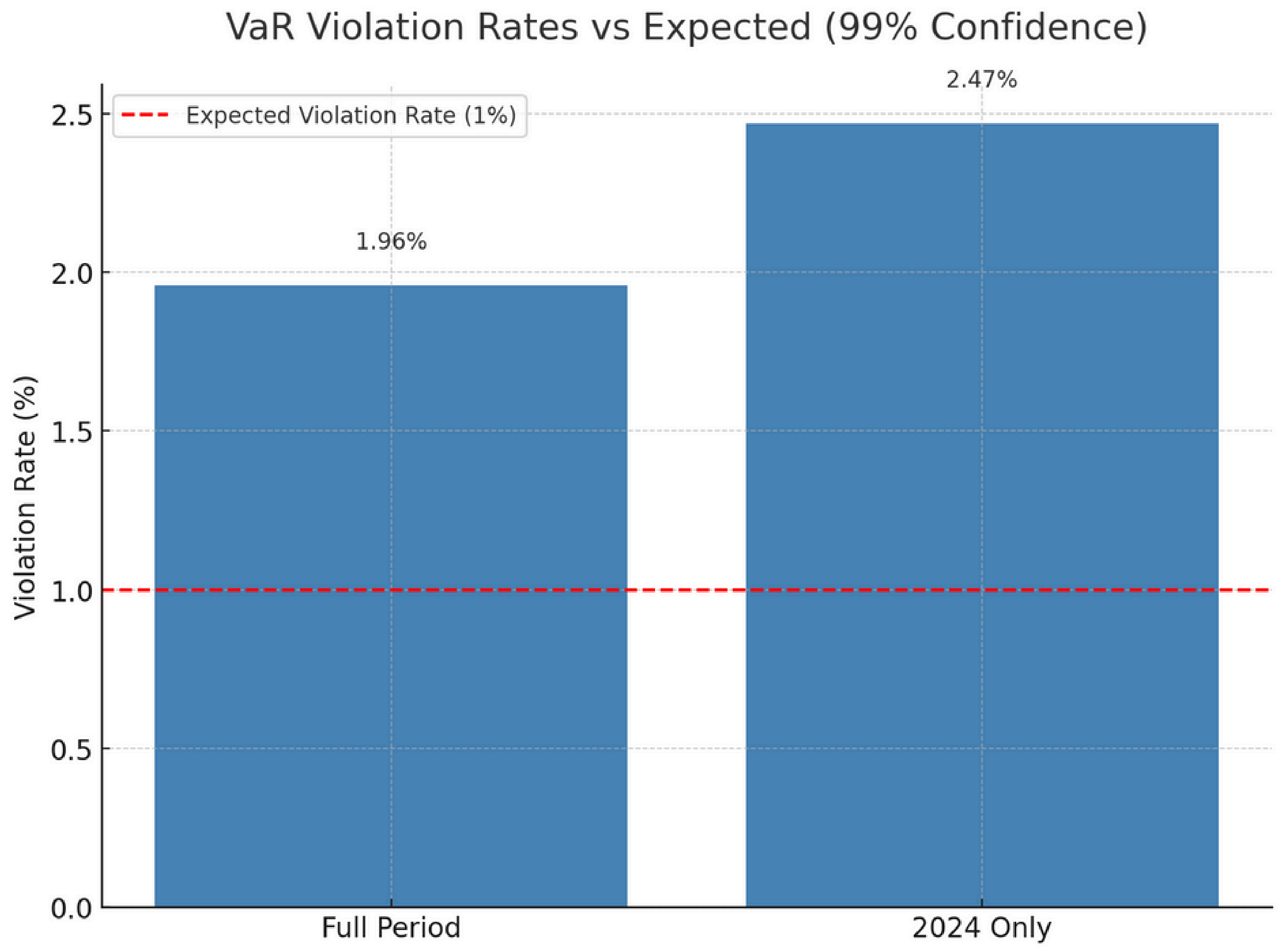




- Diversified VaR is lower than the sum of individual VaRs; shows real diversification benefit
- Assets are positively correlated, but not perfectly, allowing risk to partially offset
- The short position in NESN helps reduce overall portfolio VaR; NESN's positive correlation with SREN & NOVN creates a natural hedge

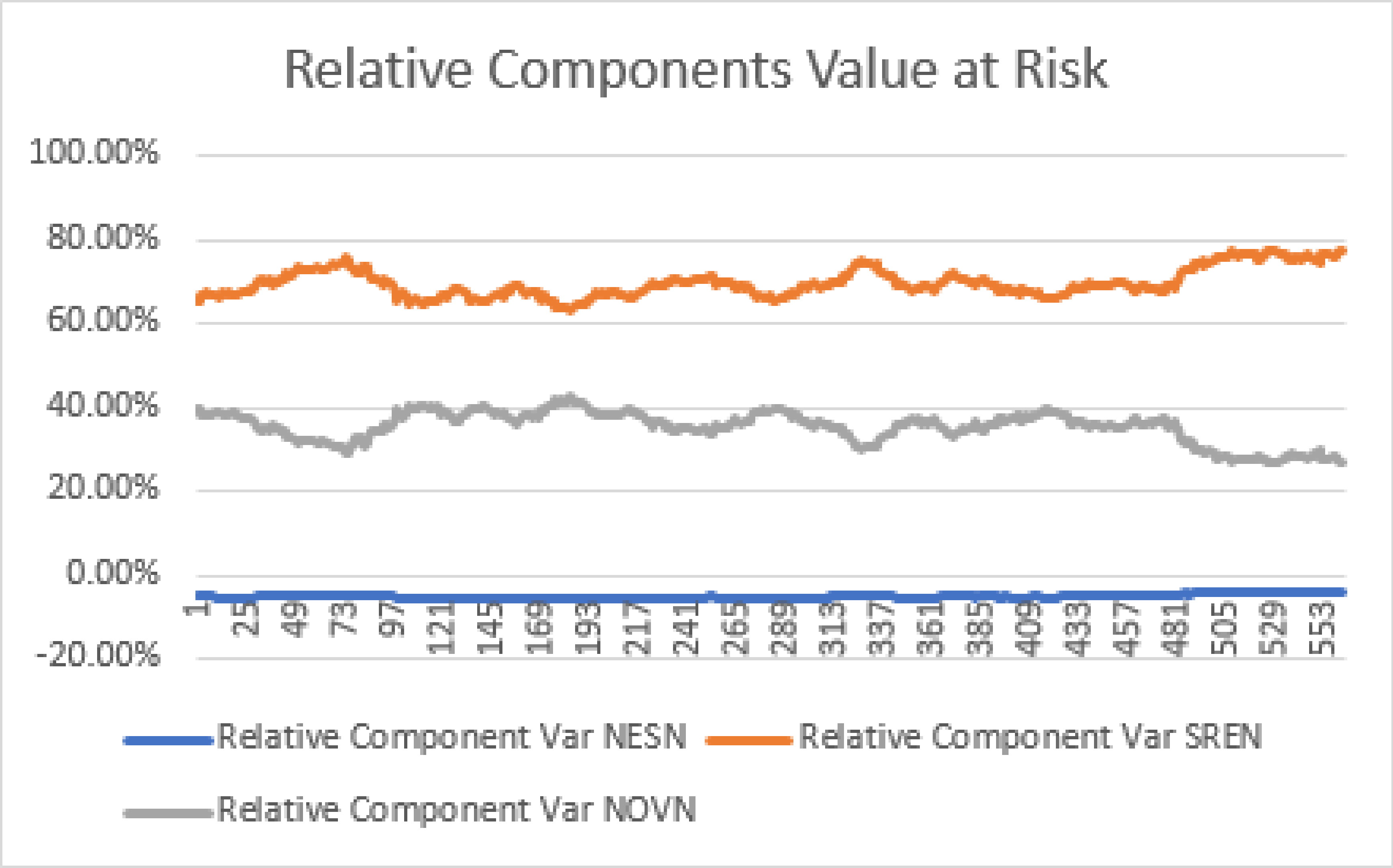
Q3: Backtesting VaR - Exceedance Frequency Over Time

- Both observed violation rates (1,97% and 2,47%) are above the expected 1% level
- Especially in 2024, market may have been more volatile or non-normal
- Suggests the AN VaR model underestimates tail risk
- Amplifies the importance of backtesting and adapting VaR models to changing conditions



Q4: CVaR and RCVaR Evolution by Position

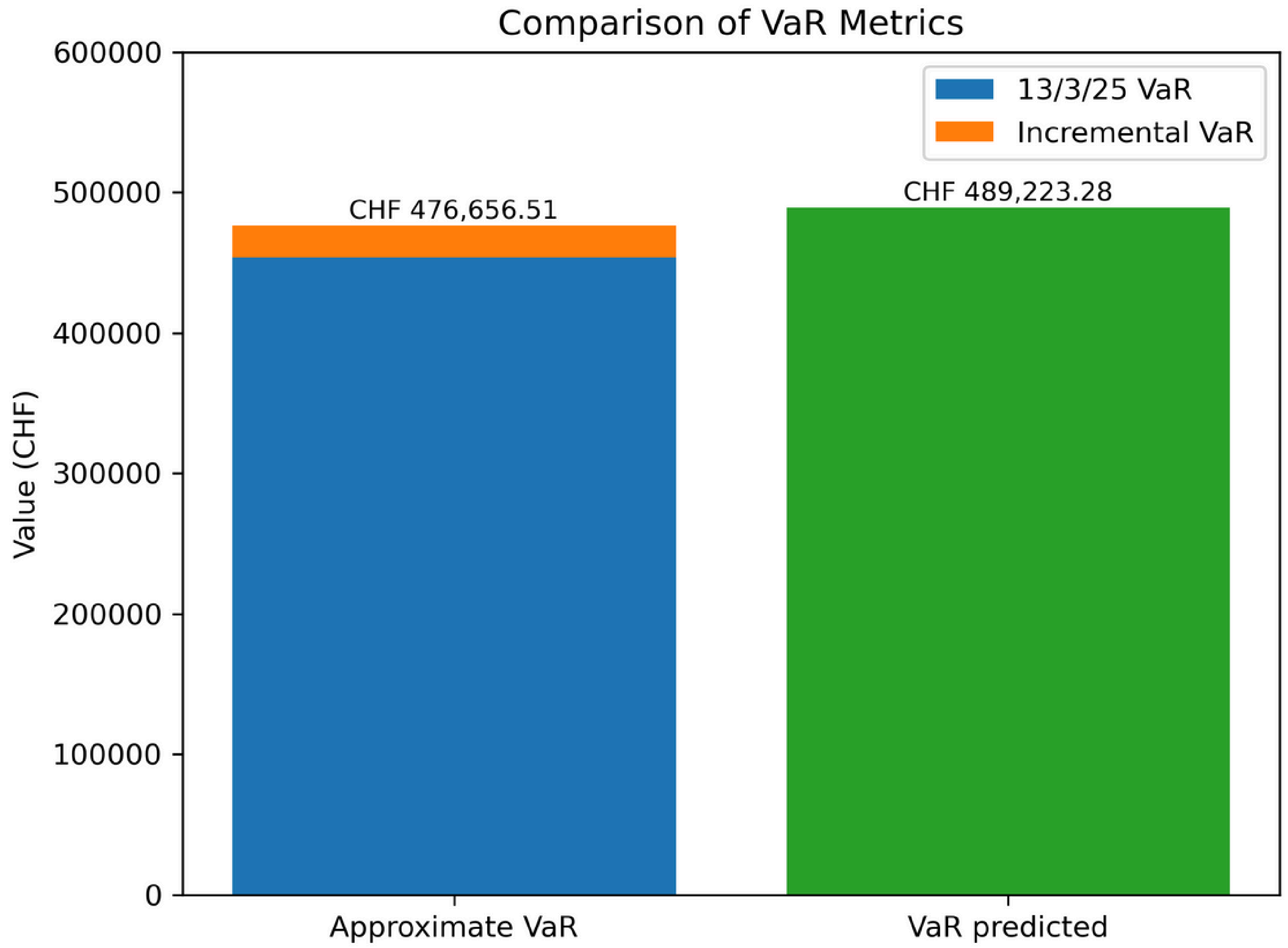
- SREN consistently drives portfolio risk, contributing ~70% of total VaR.
- NOVN's risk contribution declines over time, suggesting a changing volatility or correlation profile.
- NESN has minimal impact on portfolio VaR - less volatile.



Q5: Impact of Planned Trades on Portfolio VaR

Planned Trades:

- Buy: CHF 2,000,000 in NESN
- Buy: CHF 1,000,000 in SREN
- Sell: CHF 1,000,000 in NOVN



- The trade increased exposure to riskier assets (SREN)
- The hedging effect of NESN was reduced
- This indicates that incremental VaR can underestimate risk
- Incremental VaR is a linear approximation and may miss nonlinear impacts

Part II

Q1: Estimating Systematic Risk via Market Betas

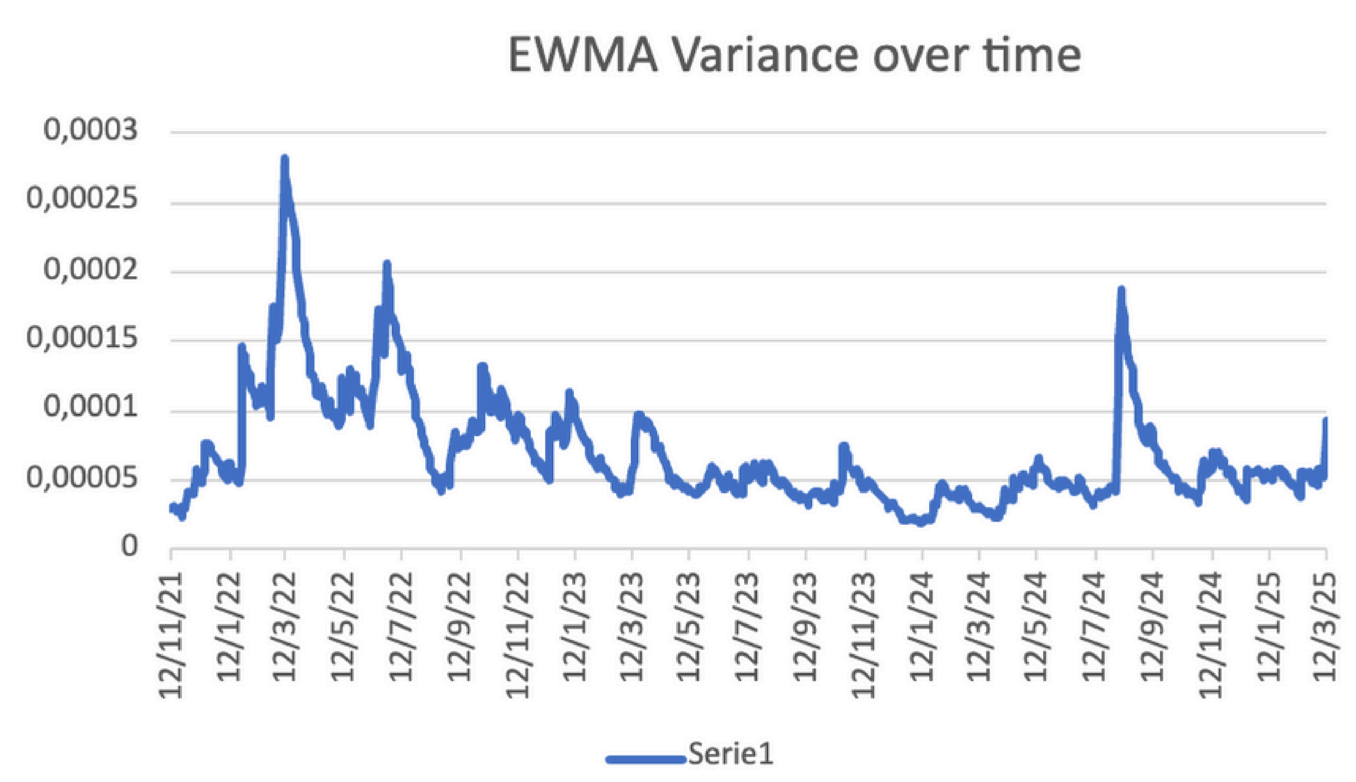
Notes:
- Beta < 1 means less responsive to market movements
- Beta > 1 means more sensitive to market movements
- Beta = 1 means 1:1 movement with the market

Less responsiveness to market means fewer returns but also less risk!

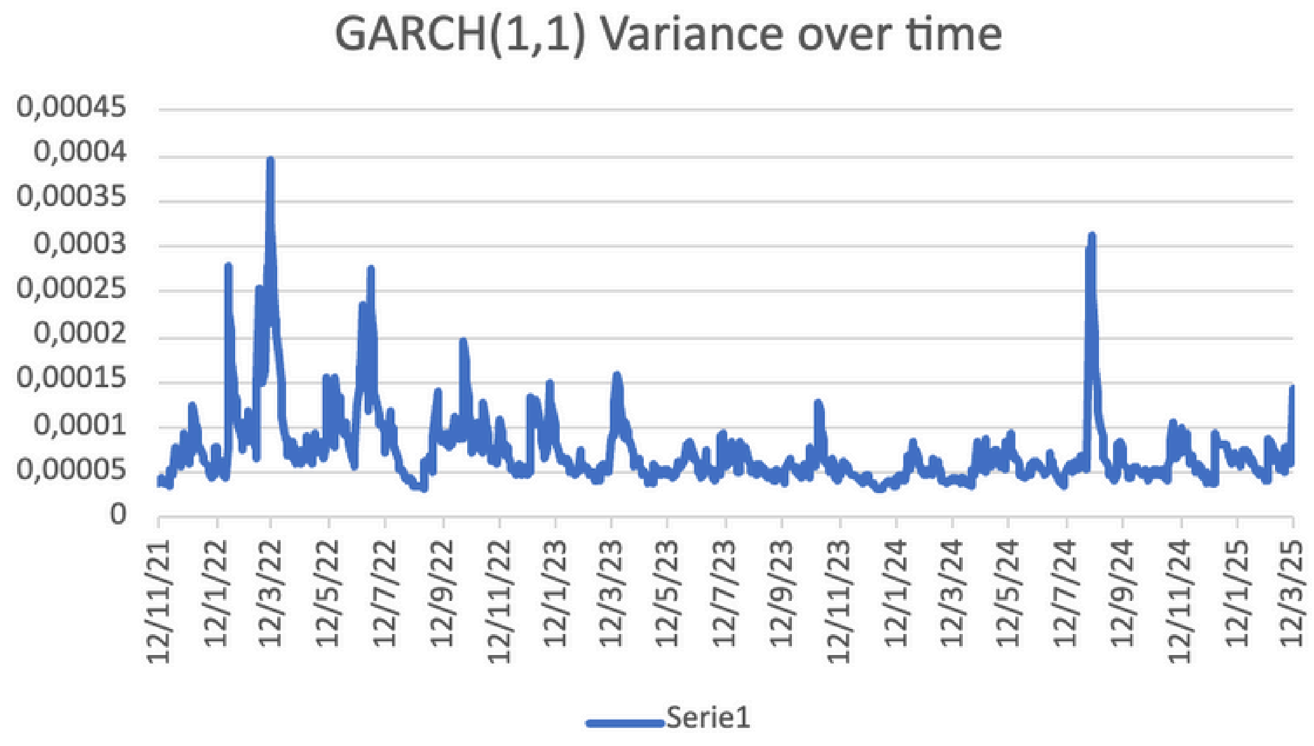
Stock	**Beta**
NESN	0.756
SREN	1.09
NOVN	0.906

Q2: Modeling Time-Varying Volatility of the SMI Index

Model	**Estimated variance**
EWMA	0.0000924961 (~9.25×10^{-5})
GARCH(1, 1)	0.000143077 (~1.43×10^{-4})



- Smoother response to shocks
- Gradual decay in variance after peak



- Strong reaction to shocks (early 2022, late 2024)
- Higher short term risk

Q3: Sharpe Diagonal VaR Comparison Across Volatility Models

Asset normal:

- Assumes constant volatilities and correlations
- Does not adapt to changing market conditions

EWMA:

- Time-varying volatility => more recent return changes get more weight, and older returns less
- More dynamic than asset normal approach

GARCH(1, 1):

- Also dynamic, but react more extremely to market shocks
- More reactive than EWMA (less smooth)

Model trusted the most: GARCH(1, 1)

Approach	**VaR(99%; 1)**
EWMA	CHF 396,544.51
GARCH(1, 1)	CHF 493,190.87
Asset normal approach	CHF 271,339.53

Thank You!